

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.87_2)

Metric

$$ds^2 = -dt \otimes dt + \cos(t) e^z dx \otimes dx + \cos(t) e^z dy \otimes dy + dz \otimes dz$$

Metric Functions

$$P = e^z$$

$$Q = \cos(t)$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2}\right) dt + \left(\frac{1}{2} \sqrt{2}\right) dz$$

$$n = \left(\frac{1}{2} \sqrt{2}\right) dt + \left(-\frac{1}{2} \sqrt{2}\right) dz$$

$$m = \left(-\frac{1}{2} \sqrt{2} \sqrt{\cos(t)} e^{\left(\frac{1}{2} z\right)}\right) dx + \left(\frac{1}{2} i \sqrt{2} \sqrt{\cos(t)} e^{\left(\frac{1}{2} z\right)}\right) dy$$

$$\bar{m} = \left(-\frac{1}{2} \sqrt{2} \sqrt{\cos(t)} e^{\left(\frac{1}{2} z\right)}\right) dx + \left(-\frac{1}{2} i \sqrt{2} \sqrt{\cos(t)} e^{\left(\frac{1}{2} z\right)}\right) dy$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{1}{2}\sqrt{2}\right) dz$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(-\frac{1}{2}\sqrt{2}\right) dz$$

$$m = \left(-\frac{\sqrt{2}e^{(-\frac{1}{2}z)}}{2\sqrt{\cos(t)}}\right) dx + \left(\frac{i\sqrt{2}e^{(-\frac{1}{2}z)}}{2\sqrt{\cos(t)}}\right) dy$$

$$\bar{m} = \left(-\frac{\sqrt{2}e^{(-\frac{1}{2}z)}}{2\sqrt{\cos(t)}}\right) dx + \left(-\frac{i\sqrt{2}e^{(-\frac{1}{2}z)}}{2\sqrt{\cos(t)}}\right) dy$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\partial_t + \frac{1}{2}\sqrt{2}\partial_z$$

$$\Delta = -\frac{1}{2}\sqrt{2}\partial_t - \frac{1}{2}\sqrt{2}\partial_z$$

$$\delta = -\frac{\sqrt{2}e^{(-\frac{1}{2}z)}}{2\sqrt{\cos(t)}}\partial_x + \frac{i\sqrt{2}e^{(-\frac{1}{2}z)}}{2\sqrt{\cos(t)}}\partial_y$$

$$\bar{\delta} = -\frac{\sqrt{2}e^{(-\frac{1}{2}z)}}{2\sqrt{\cos(t)}}\partial_x - \frac{i\sqrt{2}e^{(-\frac{1}{2}z)}}{2\sqrt{\cos(t)}}\partial_y$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}(\cos(t) + \sin(t))}{4 \cos(t)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}(\cos(t) - \sin(t))}{4 \cos(t)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{2 \cos(t) \sin(t) - 1}{8 \cos(t)^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{2 \sin(t)^2 - 1}{16 \cos(t)^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{2 \cos(t) \sin(t) + 1}{8 \cos(t)^2}$$

$$\Lambda = \frac{6 \sin(t)^2 - 7}{48 \cos(t)^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{1}{12 \cos(t)^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"